Namal University, Mianwali

Project Proposal

Project: Point-of-Sale System

Course: CSC-271 – Datebase Systems

Department: Computer Science

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Submitted To:

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1 Introduction

Fast food restaurants serve a large number of customers every day, which requires efficient management of orders. In many restaurants, Managing these orders manually can become difficult when the number of customers increases, and it may lead to delays, mistakes in orders, and difficulty in maintaining proper records.

A Point-of-Sale (POS) system helps restaurants manage this process by allowing staff to record customer orders and maintain organized records of each complete order(selected menu items, the generated bill, and the date the order was placed).

The purpose of this project is to design a Restaurant Point-of-Sale System that allows receptionists to select items from a predefined menu, generate a bill for the order, and store the order information in a database. The system will also allow admin to manage menu items, managing users account and view order history. By storing order data in a database, the system helps maintain organized order records and supports efficient restaurant operations.

2 Problem Statement

Many restaurants still take orders and manage records manually. This can cause delays, mistakes, and makes it hard to keep track of past orders. The main problems solved by the proposed system are:

- Customer orders are not recorded in a proper system, which can cause confusion.
- Menu items and prices are hard to update without a central system.
- Without the system there is no easy way to store or view past orders with their bills and dates, making it difficult for the admin to track orders.
- Without the system the admin cannot easily see all past orders or view orders by date, which makes monitoring the restaurant harder.

The proposed Point of Sale system aims to solve these problems by providing an organized database system for managing customer orders, generating bills, and allowing administrative control.

3 Objectives

The main objectives of the proposed system are:

1. To design a database system for managing restaurant orders.
2. To allow receptionists to create and manage customer orders efficiently.
3. To automatically generate customer bills based on selected menu items.
4. To allow administrators to manage and update menu items.
5. To allow administrators to add or delete system users.
6. To maintain a record of all orders.
7. To allow administrators to view order details within a specific date range.

4 Scope

The proposed Restaurant Point-of-Sale System will include the following features:

- **Menu Display:** The system will provide a predefined menu that contains all available food items along with their prices. The receptionist will be able to view the menu and select items while creating a customer order.
- **Order Management:** The receptionist will be able to create a new order for each customer. Items can be added, removed, or updated before the order is finalized.
- **Bill Generation:** The system will generate a bill based on the selected menu items and their quantities. When the receptionist completes the order and performs checkout the bill along with the order items and date will be saved in the system for order history.
- **Menu Management:** The admin will be able to add new menu items, remove existing items, and update item prices .
- **User Management:** The admin will have the ability to add or remove system users.
- **Order History Management:** The system will store all order records in the database, allowing the administrator to view and filter orders using a specific date range.

The waiter will only serve the prepared food and will not interact with the system.

5 Data Collection

Data required to test the system will be collected as follows:

- Menu data: obtained from online available restaurant menus and publicly available menu sources.
- User Credential data (admin and receptionist): manually created to simulate system users.
- Order data: generated by creating sample customer orders during system testing.
- Bill data: automatically generated by the system based on selected menu items.

6 Technology Stack

The following technologies have been selected based on their suitability for a relational database-driven Restaurant Point-of-Sale system:

Layer	**Technology**	**Purpose**
Database	MySQL 8.4+	Core relational database for storing and managing menu items, orders, and user accounts.
Backend / Program Logic	C++	Implements the main logic of system.
Frontend	Raylib (C++)	Provides the graphical user interface for receptionists and admin to interact with the system.
Authentication	Built in login system	Manages secure access for admin and receptionist users.
Version Control	Git / GitHub	Source code version control and collaboration among team members.

7 System Functionality

The proposed system will provide the following functionalities:

1. Login access for users.
2. Viewing the predefined restaurant menu.

3. Creating and managing customer orders.
4. Adding menu items to an order.
5. Removing items from an order.
6. Updating quantities of items in an order.
7. Generating a bill for each order and saving it with order details and date.
8. Admin ability to add new menu items.
9. Admin ability to update menu items and prices.
10. Admin ability to remove menu items.
11. Admin ability to add or remove system users.
12. Admin ability to view order history.
13. Admin ability to filter order records by date.

8 User Classes

The system defines two main user roles, each with distinct access levels within the system:

Role	Description	Access Level
Admin	Manages the entire system, including menu and users	Full access: add/remove/update menu items, add/remove users, view order history, filter orders by date
Receptionist	Handles customer orders at the counter	Access to view menu, create/manage orders, generate bills

9 Task Division

The project has been divided among team members to ensure balanced contribution and responsibilities.

Phase	Muhammad Shoaib	Allah Ditta
Documentation	Section: Introduction, Problem Statement, Objectives, Scope, Data Collection, Technology Stack	Section: System Functionality, User Classes, Task Division, Conclusion, References, AI Prompts
Database Design	ER Diagram; Tables for Orders, Menu Items, Users, Bills	ER Diagram; Tables for Orders, Menu Items, Users, Bills
Backend	ERP functionalities for admin users; C++ logic for admin operations	ERP functionalities for receptionist users; C++ logic for order management
Frontend	ERP functionalities for admin user interface	GUI using Raylib for receptionists and admin; Integration with backend and database

10 Conclusion

The proposed Point of Sale System helps to manage customer orders and store order history in an organized way. Receptionists can create and update orders from a predefined menu and generate bills, which are saved with order details and dates. Admins can manage menu items, add or remove users, and view order history by date. Using a database ensures that all orders, menu items, and user information are stored accurately and can be easily accessed and making restaurant operations more easy.

11 References

1. OpenAI. (2026). *ChatGPT* [Large language model]. OpenAI. Available at: `https://chat.openai.com/`

Appendix 1: AI Tools Used

Tool	Purpose	Full Prompt	Accessed Date
ChatGPT	Help to adjust our Software Engineering project to a smaller scope for Database Systems	"This is the SRS of our Software Engineering project. Our instructor reduced the project for the Database Systems course. Some features like order status and actor like waiter, cashier and kitchen staff is removed and one actor is added receptionist."	10 March 2026
ChatGPT	Help to choose the technology stack for the POS system	"I am making a Database Systems proposal for a Point of Sale system. Suggest a suitable technology stack including database, backend, frontend, and other tools.."	11 March 2026
ChatGPT	Help to write a professional citation for ChatGPT in References	"Write professional citation of Chat GPT in proper format to include in the References section."	12 March 2026
ChatGPT	Help to format the LaTeX report and reduce extra spaces	"Add the project name at the top right of each page. Reduce extra vertical space between sections and in the table of contents."	12 March 2026